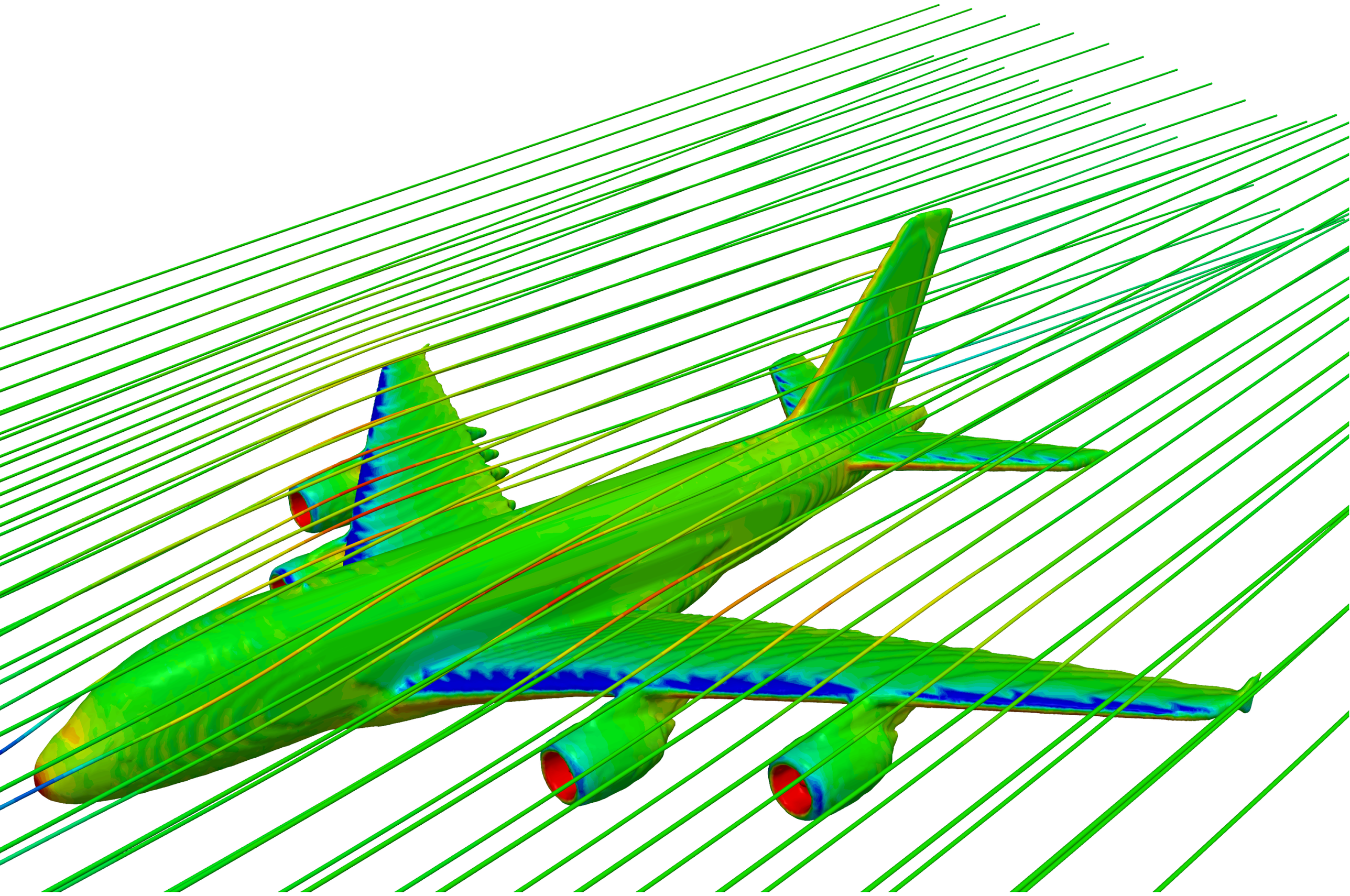
How a 500-tonne plane stays up.

Air speeds up over the top of the wing, so its pressure drops (the blue band) and the wing is pulled upward. We solved this airflow around a full-size Airbus A380 at nine angles, using the Navier–Stokes equations.



$$\text{fuel} \propto 1 \div (L/D)$$

Surface pressure

Low: suction — High: air hits head-on

Air paths, coloured by speed

312 km/h — 422 km/h

More lift for each unit of drag means less fuel.

In steady flight, lift equals the plane's weight, so drag = weight ÷ (L/D). The angle with the highest lift-to-drag ratio burns the least fuel and puts out the least CO_2. Turn the knob on the screen to tilt the wing.

SDG 9

Industry, innovation and infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialisation, and foster innovation.

Solved with OpenFOAM, k-ω SST · wind 367 km/h, sea-level air

NavFlow · NPS International School STEM Exhibition